

Exercise A (KNN)

Let $y_i = f(x_i) + \varepsilon_i$, and let $\hat{f}(x) = \frac{1}{k} \sum_{i \in N_k(x)} y_i$. Then, the bias of the KNN estimator is

$$E[\hat{f}(x)] = \frac{1}{k} \sum_{i \in N_k(x)} E[y_i] = \frac{1}{k} \sum_{i \in N_k(x)} f(x_i)$$

Hence, the bias is

$$\text{Bias}\{\hat{f}(x)\} = \frac{1}{k} \sum_{i \in N_k(x)} f(x_i) - f(x),$$

whereas the variance is

$$\text{var}\{\hat{f}(x)\} = \text{var}\left(\frac{1}{k} \sum_{i \in N_k(x)} y_i\right) = \frac{1}{k^2} \sum_{i \in N_k(x)} \text{var}(y_i) = \frac{k\sigma^2}{k^2} = \frac{\sigma^2}{k}$$

independent r.v.!

The equivalent degrees of freedom are:

$$df = \frac{1}{\sigma^2} \sum_{i=1}^m \text{cov}(\hat{f}(x_i), y_i) = \frac{1}{\sigma^2} \sum_{i=1}^m \text{cov}\left(\frac{1}{k} \sum_{e \in N_k(x_i)} y_e, y_i\right)$$

$$= \frac{1}{\sigma^2 k} \sum_{i=1}^m \sum_{e \in N_k(x_i)} \text{cov}(y_e, y_i)$$

properties of the covariance

$$= \frac{1}{\sigma^2 k} \sum_{i=1}^m \sigma^2 = \frac{m}{k} \frac{\cancel{\sigma^2}}{\cancel{\sigma^2}}$$

$$= \frac{m}{k}$$

Now note that

$$\text{cov}(y_e, y_i) = 0 \quad \text{if } e \neq i$$

$$\text{cov}(y_e, y_i) = \sigma^2 \quad \text{if } e = i.$$

$$\Rightarrow \sum_{e \in N_k(x_i)} \text{cov}(y_e, y_i) = 0 + \dots + \sigma^2 + \dots + 0 = \sigma^2$$